Guarded Micro-UAV
Three-Tier Parts Reference for CAD & FEA

Cheapest (Value) • Most Expensive (Capability Ceiling) • Most Efficient (Mission-Optimal)

Snapshot 2026-06-22. Prices and masses are catalog values or engineering allowances and remain **planning evidence** until delivered parts are weighed and bench-tested. Do not freeze maximum mass, guard geometry, or load cases from this sheet alone — the values marked BENCH_REQUIRED must be measured.

Generated from `Design Report/BOM.csv` and `unrestricted-tier-study-2026/DECISION.md`.

1. Tier Comparison by Subsystem

Pick one column to model. The Value column is the current locked Stage 1 build (and FEA baseline). The Mission-optimal column is the recommended target *after* the propulsion thrust gate passes. The Capability-ceiling column is a reference only — it is not hand-throwable and breaks the indoor release experiment.

Subsystem	Cheapest (Value)	Most Expensive (Ceiling)	Most Efficient (Optimal)
Frame & guards	Custom 2" nylon frame	Custom 23" guarded industrial quad	GEPRC GEP-CL35 V2 (142 mm, 3.5")
Flight controller	Kakute H7 Mini v1.5	Pixhawk 6X Pro + PM02D	Pixhawk 6C Mini + PM02 V3
ESC / distribution	HGLRC XJB BS13A 2-3S	4x A6-M FOC/CAN arms	Tekko32 F4 Mini 50A AM32
Motors	4x Happymodel EX1103 11000KV	4x T-Motor A6-M KV280	4x GEPRC SPEEDX2 2105.5 2650KV
Propellers	Gemfan Hurricane 2023-3	MF2311P-M 23"	Gemfan D90-3 ducted
Battery	GNB 2S 550 mAh 100C	Large 6S (sized post-test)	6S 1050-1300 mAh high-rate
Companion compute	Arduino Nicla Vision (MCU)	Jetson AGX Orin Industrial 64 GB	Raspberry Pi Zero 2 W
Vision / camera	Nicla onboard GC2145	Luxonis OAK 4 D Pro W	Luxonis OAK-D Lite
Flow / range	Matek 3901-L0X (MSP)	ARK Flow + LightWare SF45/B	ARK Flow (DroneCAN)
Manual control RX	BetaFPV ELRS Lite	CubePilot Herelink 1.1	RadioMaster RP3 V2 diversity
Telemetry / logs	CRSF + onboard log	Herelink + Microhard P900	SiK V3 100 mW pair + microSD
Perception power	FC rail (if measured safe)	Custom isolated DC/DC	Separate 5 V regulator
Flying-HW cost	~$333	>$6,200 (>$9,000 complete)	~$1,030
Vehicle scale	~129.8 g nominal	Multi-kilogram	~0.55-0.70 kg (bench-frozen)
Build decision	First build & control case	Reference only - do not build	Recommended after thrust gate

2. Cheapest — Value Build (locked Stage 1, FEA baseline)

~$333 flying hardware, ~129.8 g nominal. Rows shaded red are EOL/legacy and must be re-confirmed or substituted before ordering (see Open Questions OQ-007, OQ-008).

Subsystem	Part	Qty	Mass nom (g)	Cost nom ($)	Notes
Flight control	Holybro Kakute H7 Mini v1.5	1	5.5	65	v1.5 effectively unavailable since ~mid-2025
ESC	HGLRC XJB BS13A 2-3S 4-in-1	1	3.0	18	20x20; legacy, mostly sold in flytower
Motors	Happymodel EX1103 11000KV	4	15.2 ea	36	121.9 gf @ 9.2 A vendor pt; bench @ 7.0 V
Propellers	Gemfan Hurricane 2023-3	1 set	3.5	6	1.5 mm T-mount, 4x 0.88 g
Battery	GNB 2S 550 mAh 100C LiHV XT30	1	29.0	14	7.6 V LiHV; sag BENCH_REQUIRED
Control link	BetaFPV ELRS Lite RX	1	0.46	12	5 V ELRS/CRSF on SERIAL6
Status	VIFLY Finder Mini + WS2812	1 set	3.3	18	self-powered buzzer + LED
Vision	Arduino Nicla Vision ABX00051	1	19.8	90	MCU+cam+IMU+ToF; rolling shutter
Positioning	Matek 3901-L0X flow + lidar	1	2.0	26	EOL; flow+range over 1 MSP UART
Electrical	XT30 + 22/26 AWG + JST-SH	1 set	5.5	8	weigh completed harness
Mechanical	Printed nylon frame + guards + M2	1 set	42.5	40	CAD/FEA/push test required

Support (non-flying): ToolkitRC M6AC charger ~$60, PVC+net cage ~$40, tethered release rig ~$20, DIY 1-2 kg thrust stand ~$100. All-in project ~$350-650.

3. Most Efficient — Mission-Optimal Build (~$1,030, ~0.55-0.70 kg)

Recommended target after the guarded propulsion thrust gate passes. Buy the propulsion-validation subset first (one motor, prop set, 50 A ESC, 6S pack, one guard) and require ≥350 gf/motor at minimum loaded voltage before committing the avionics.

Subsystem	Part	Cost ($)	Notes
Frame & guards	GEPRC GEP-CL35 V2 3.5"	69.99	142 mm, 95 mm guard ID, 133.7 g frame
Flight controller	Pixhawk 6C Mini + PM02 V3	149.98	dual isolated/heated IMU, H743
ESC	Holybro Tekko32 F4 Mini 50A AM32	~69.99	20 mm, 6S, inferred standalone price
Motors	4x GEPRC SPEEDX2 2105.5 2650KV	~88	21 g ea, 12x12 M2 mount, 2-6S
Propellers	Gemfan D90-3 ducted tri-blade	incl.	90 mm, 2.3 g, for 3.5" ducts
Battery	6S 1050-1300 mAh high-rate LiPo	~45	final capacity bench-selected
Companion	Raspberry Pi Zero 2 W	15	MAVLink/DepthAI host
Vision	Luxonis OAK-D Lite	169	global-shutter stereo + on-device NN
Flow / range	ARK Flow (DroneCAN)	250	5 g, PAW3902 + ToF + IMU on CAN
Control link	RadioMaster RP3 V2 diversity	19.99	TCXO, dual-antenna ELRS
Telemetry	Holybro SiK V3 100 mW pair	58.99	independent MAVLink + onboard log
Perception power	Separate 5 V regulator	20.99	isolates OAK/Pi transients from FC
Mounts / wiring / status	allowance	~68	adapter plate, isolated avionics deck

The GEP frame does not natively accept the Pixhawk 6C Mini footprint — a stiff adapter plate and isolated avionics deck are an FEA item. Excludes the optional $1,075 Tyto thrust stand (support gear).

4. Most Expensive — Capability Ceiling (reference only)

Highest rational COTS capability, independent of cost: >$6,200 airborne subtotal, likely >$9,000 complete. **Do not build for the release mission** — multi-kilogram scale makes hand release unsafe and changes the project into an industrial UAV program. Listed so the design space is bounded.

Subsystem	Part	Cost ($)
Flight controller	Pixhawk 6X Pro Standard v2B + PM02D	797.99
Companion	NVIDIA Jetson AGX Orin Industrial 64 GB	2,899
Vision	Luxonis OAK 4 D Pro W	1,049
Propulsion	4x T-Motor A6-M 6S FOC modules + 23" props	~400+/arm
Flow / range	ARK Flow + LightWare SF45/B scanning lidar	—
Control / telemetry	CubePilot Herelink 1.1 + Microhard P900 pair	449 (P900 pair)
Test gear	Tyto Robotics Series 1585 thrust-stand bundle	1,075
Frame	Custom carbon/composite guarded industrial quad	custom

5. Mechanical & Geometry Data for CAD / FEA

Use these for the structural model and load cases. Guard deflection and stress are the governing FEA outputs; all are BENCH_REQUIRED to confirm.

Masses (point/distributed loads)

Item	Mass
Value frame + guards + M2 (allowance)	best 35.0 / nom 42.5 / worst 50.0 g
Value flying mass (Sigma)	best 117.96 / nom 129.76 / worst 149.5 g
Value unmodeled-HW allowance	5.0 g
Value proposed frozen max (NOT frozen)	155.0 g; abort threshold 225 g
EX1103 motor (Value)	15.2 g each (x4 as corner point loads)
GEP-CL35 V2 frame (Optimal)	133.7 g bare frame
SPEEDX2 2105.5 motor (Optimal)	21 g each
D90-3 prop (Optimal)	2.3 g each
Optimal vehicle target	~0.55-0.70 kg, bench-frozen

Mounting geometry

Feature	Spec
Value FC/ESC stack	20 x 20 mm pattern
Optimal frame	142 mm motor-to-motor, 95 mm guard inside diameter, 3.5" props
Optimal FC holes	25.5 x 25.5 mm
Optimal motor holes	12 x 12 mm (M2), matches SPEEDX2 2105.5
Optimal battery mount	integrated XT60, 6S current
6C Mini adapter	frame does NOT natively fit 6C Mini; stiff adapter plate + isolated avionics deck required

Load cases & acceptance gates (structural)

Load case / gate	Acceptance criterion
Guard lateral deflection	≤ 1.0 mm at the defined proof load
Guard permanent set	none after release-envelope drop test
Prop-to-guard clearance	no contact under load (verify duct interaction)
Thrust-to-weight (Value)	≥ 112.5 gf/motor @ 7.0 V for full 225 g reserve
Thrust-to-weight (Optimal)	T/W ≥ 2.0 at min loaded voltage; ≥ 350 gf/motor @ 700 g screen
Thrust rise (10-90%)	≤ 100 ms
Battery sag at acceptance step	≤ 10%

Modeling note: guards alter propeller thrust and transient response; model the guard-to-frame joint as the critical region. Freeze the maximum flying mass only after the complete CAD mass model exists and receipt masses are measured (Open Question OQ-002).